# RAJ MAURYA

**B.Tech Mathematics & Computing, IIT Mandi** rajmahimaurya@gmail.com

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## EXPERIENCE

**Software Engineering Intern**, *Pinch (Addpinch Technologies)* May'26–Jul'26
- Migrated guest-list and event-chat interfaces; built reusable React Native category components
- Fixed iOS state updates for AI-description editing and implemented single-attendee frontend logic
- Tested the app end to end, documented reproducible bugs, and delivered changes through reviewed pull requests

**Teaching Assistant**, *Himshikhar, IIT Mandi (part-time)* May'26–Jul'26
- Mentored undergraduate students and explained course concepts
- Supported course delivery and assessment

**Research Intern**, *IIT Mandi* Dec'24–Jan'25
- Studied NP-completeness and formal reductions; formulated and encoded selected problems

## RESEARCH

**SAGE-PINN: Nanofluid Flow & Heat Transfer in a Stenosed Artery** arXiv

PyTorch · COMSOL · NumPy · LaTeX | First author, under review, Journal of Computational Physics
- Singularity-free, geometry-encoded PINN for nanofluid flow and heat transfer; compared against 40 COMSOL cases

**SAFE-PINN: Electroosmotic Flow in Cylindrical Nanochannels** Manuscript on request

PyTorch · COMSOL · SciPy · LaTeX | First author, under review, Physics of Fluids (Special Issue)
- Coupled flow, ion transport and electric potential with exact boundary formulations; 32-case parameter sweeps against reference results

**Deep Neural Superhedging** Repo on request

PyTorch · NumPy · SciPy · Monte Carlo | Ongoing
- Independent replication of neural superhedging with LSTM policies, cash calibration and capacity controls; bounded empirical extension completed

**Paper-Devices: Soil Analyzer Research & Bench Platform** Repo on request

React · TypeScript · Express · PostgreSQL · PWA | Ongoing, with a faculty member
- Paper-based smartphone soil-analyzer research; chemistry and device validation are still at the research stage
- Built Bench, a lab-inventory PWA with stock history, QR labels, exports and assisted intake

## PROJECTS

**JourneySync — AMEX CodeStreet 2026 Finalist (Top 6)** Git

FastAPI · Next.js · PostgreSQL · MongoDB · Redis
- Cross-channel identity resolution and customer-journey stitching with confidence scoring and audit records
- Review workflow for unresolved cases and live analytics over stitched journeys

**Neuro-Trade — PINN for Indian-Equity Dynamics** Git

PyTorch · SciPy · KDE · pandas
- GBM and Heston/Fokker–Planck solvers, empirical density estimation, and a PINN that learns drift and diffusion (PINN phase in progress)

**Deepfake Video Detection** Git

PyTorch · dlib · OpenCV · scikit-learn
- Unsupervised detector using facial/texture cues as pseudo-labels, enhanced contrastive learning and frame-to-video aggregation

**NetSentinel — Network Threat-Monitoring Dashboard** Git

FastAPI · React · Scapy · WebSockets
- Live packet capture, PCAP analysis/replay, behavioural and fingerprint-based detectors, alerts and historical views

**Byte-Chat — Campus Community Chat** Git

Next.js · TypeScript · Express · Socket.IO · PostgreSQL · Redis
- Anonymous and group messaging, notifications, moderation/reporting and Redis-backed real-time delivery

**C-TAEA — Constrained Multi-Objective Optimization** Git

Python · NumPy
- Independent implementation of the IEEE TEVC algorithm, benchmarked against peer evolutionary algorithms with IGD and hypervolume

**GenWeb — AI Website Builder** Git Web

Next.js · Supabase · Clerk · Octokit
- Prompt-driven site generation with live previews, project storage and GitHub upload; I built the backend and model integration

## SUMMARY

Final-year Mathematics & Computing undergraduate working on research and software side by side: physics + AI/ML (two PINN papers under review), full-stack systems, and quantum computing on the side.

## EDUCATION

**B.Tech, Mathematics & Computing**

*IIT Mandi · Aug'23–May'27 (expected)*
- **CGPA:** 8.73/10 (through Sem VI)
- **Coursework:** Design of Algorithms, Data Structures, Databases, Software Engineering, Computer Networks, Computer Organization, Deep Learning, Data Science, Data Mining, Probability & Statistics, Linear Algebra, Mathematical Programming, Numerical Analysis, Graph Theory, PDEs/ODEs, Quant Finance. *Ongoing:* Financial Engineering

## TECHNICAL SKILLS

**Languages**
Python, C++, JavaScript, TypeScript, SQL, Bash

**ML / Data**
PyTorch, TensorFlow/Keras, scikit-learn, OpenCV, NumPy, pandas, SciPy

**Backend / Web**
FastAPI, Node.js/Express, React, Next.js, Socket.IO, REST, WebSockets

**Data / Tools**
PostgreSQL, MongoDB, Redis, Supabase, Git, Linux

**Scientific / Quant**
PINNs, ODE/PDE/SDE modelling, Monte Carlo, optimisation, evolutionary algorithms, COMSOL, MATLAB, LaTeX

## KEY ACHIEVEMENTS

- **AMEX CodeStreet 2026 Finalist:** top 6 team, customer-journey stitching platform
- **Two PINN papers under review:** J. Computational Physics; Physics of Fluids

## MORE PROJECTS

- **MovieFlix:** ticket booking and theatre admin; React, Express, MongoDB, Razorpay (Git)
- **YT-WrapUp:** 3D exploration of 28,000+ YouTube records (embeddings, UMAP)
- **NFL Big Data Bowl 2027 (Kaggle, submitted):** Combine measurements vs. game movement and outcomes; Polars, DuckDB (on request)
- **Trade:** YOLO/CNN chart-pattern research and NSE screener (on request)
- **EXOCORTEX:** personal knowledge and learning-history platform (on request)
- **Portable Oxygen Concentrator:** IoT prototype for low-resource settings

## CERTIFICATIONS & OTHER

- Machine Learning, Coursera (Andrew Ng)
- Quantum Computing, Coursera
- Dynamic Programming, Coursera
- Member, Research Society, IIT Mandi (since 2024)